

QIRUI MIAO

Graduate Technical Candidate | Materials & Data | Product, Applications & Manufacturing

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PROFILE

UCL MSc Advanced Materials Science candidate (Data-driven Innovation) with a KCL background in geography, environmental science and spatial data science. Cross-disciplinary evidence spans computational materials, scientific model validation, optics manufacturing and technical sales, CAD/FDM prototyping, spatial analysis and sustainability research. Three summer placements at XLASER combined customer-facing work with production, optical-coating, QC and operations exposure.

EDUCATION

University College London (UCL) — MSc Advanced Materials Science, Data-driven Innovation | Sep 2025 –

Expected Sep 2026

- Dissertation submitted: DFT-Derived Oxidation Energy Prediction for Organic Cathode Molecules Using Machine Learning — a Gaussian-to-RDKit benchmark across four organic cathode families.
- Selected published module marks: Advanced Materials Processing & Manufacturing 74.81; Integrated Data-Driven Materials Science & Digitalisation 73.17; Literature Project 70.00.

King's College London (KCL) — BSc (Hons) Geography and Environmental Science (with Spatial Data Science) |

Sep 2022 – Jul 2025

- Dissertation: Optimizing the UK's Offshore Wind Energy Permitting Process: Lessons from Denmark, Germany, and the Netherlands — comparative case-study, regulatory-document and policy analysis.

INDUSTRY EXPERIENCE

XLASER OPTICS CO., LTD., Shanghai — Technical Operations & Sales Intern | Jun–Sep 2023 | Jun–Sep 2024 | Jun–

Sep 2025

- Worked across international sales, production, optical-coating workshop operations, QC, packing/shipping, inventory/administration and ad-hoc IT support in a manufacturer of laser optics, precision optical components, optical coatings and laser-related components.
- Read and modified customer drawings and created 10+ 2D/3D AutoCAD designs; independently took laser-protective-panel jobs from dimensional requirements through CAD, laser cutting/marking, QC and shipment preparation.
- Assisted coating-machine loading and operated interfaces under supervision, including basic settings related to vacuum/pressure, film thickness, deposition time, temperature, optical monitoring, recipe selection and coating materials.
- Performed dimensional and surface QC using micrometers, vernier calipers and high-intensity-light inspection for thickness, dimensions, scratches/pits, edge defects and coating defects.
- Represented the company at CIOE (2023–25), LASER World of PHOTONICS CHINA (2024–25) and CIIF (2024–25), engaging 100+ visitors per exhibition (~20% international), handling product demonstrations, technical questions, lead capture, email follow-up and quotations; some leads progressed to orders.
- Collected shop-floor workload data and helped design an Excel-based piece-rate framework for approximately 20–30 production employees, incorporating process complexity, wage safeguards and incentives for learning higher-skill operations.

RESEARCH & TECHNICAL PROJECTS

MSc Dissertation — Organic Cathode DFT/ML Workflow Validation | UCL | 2026

- Evaluated a supplied Gaussian-to-RDKit benchmark covering 840 molecules across phenothiazine, phenoxazine, triarylamine and carbazole families; 827 records entered a 231-feature modelling stage comparing XGBoost, Random Forest, SVR and Linear Regression.
- Independently audited the implemented random 80/20 validation against the intended scaffold-holdout design, reframing XGBoost performance (R^2 0.963; MAE 0.045 eV; RMSE 0.063 eV) as an interpolation baseline rather than unseen-scaffold generalisation.
- Checked the Gaussian campaign against 3,432 expected jobs (3,431 logs; 3,382 Normal terminations), verified the Eoxidised – Eneutral Hartree-to-eV target definition, and distinguished genuine Random Forest SHAP from permutation-importance fallbacks.

Machine Learning for Computed Band-Gap Prediction | UCL | Python, Jupyter, XGBoost, RF, SVR, ANN

- Built and evaluated a descriptor-based regression workflow for a 1,306-row double-perovskite dataset, with leakage screening, feature reduction and criterion-aware model comparison.

LLZO Molecular Dynamics & Cu Vacancy DFT | UCL | LAMMPS, ASE, Quantum ESPRESSO

- Ran 80,000-step NPT LLZO simulations at 500/1100 K with block-averaged uncertainty; separately calculated an unrelaxed fcc Cu vacancy formation energy of 0.93 eV using consistent PBE/PAW SCF calculations.

Personal Product Design & Additive Manufacturing | Fusion 360, Bambu Studio, FDM

- Modified a MagSafe stand into an integrated iPhone–Apple Watch–AirPods charging station with a retractable Watch-charger mechanism; iterated geometry, sliding clearances and print parameters to resolve warping and print failures, producing a daily-use final design.

Spatial & Environmental Data Projects | KCL | ArcGIS, raster GIS, census/land-cover data

- Analysed London car-ownership change from 1991–2021 using UK Census data, ArcGIS spatial joins and cartographic comparison; separately mapped Scotland land-cover change (1990/2000/2015) using raster cropping, classification, change detection and quantitative area comparison.

LEADERSHIP & STAKEHOLDER EXPERIENCE

UCL Hiking Club — Walk Leader / Committee Activity Support

- Co-organised a 400+ participant White Cliffs day hike, contributing to route/transport planning, ticketing, group segmentation, leader allocation, participant communication and on-the-day coordination.
- Co-organised a five-day Yorkshire Three Peaks residential for ~60 participants, including hostel/coach logistics, registrations, dietary requirements, group meals and multiple route options for different ability levels.

UCL Materials Programme — Academic Representative | 2025–26

- Acted as a communication bridge for the cohort, including Chinese-speaking students, and represented feedback to staff on assessment, scheduling and NSCI0017 research-project organisation.

TECHNICAL SKILLS

Data & scientific computing: Python, pandas, NumPy, Matplotlib, XGBoost, Jupyter, Git/GitHub; Materials Project API/mp-api; model evaluation and scientific data interpretation.

Computational materials: ASE, Quantum ESPRESSO, LAMMPS; Gaussian workflow/log QC and output inspection; UCL Myriad HPC file/project management.

CAD, manufacturing & prototyping: AutoCAD (2D/3D), Fusion 360, Bambu Studio, FDM 3D printing (PLA/PETG), laser cutting/marking, optical-coating workshop exposure, micrometer/vernier-caliper and surface QC.

Spatial/environmental: ArcGIS, QGIS, R, Google Earth Engine, remote sensing, spatial analysis and data visualisation.

Languages: Mandarin Chinese (native); English used throughout UK university study, technical writing, international exhibitions and customer communication.